

CHAPTER 4: POWER AND REPLICATION— DESIGNING POWERFUL EXPERIMENTS

Michael D. Casler

Power is the probability of correctly rejecting a null hypothesis that two or more treatment means are equal to each other when in fact they are different. Designing experiments with high power is critical for detecting small, but biologically meaningful, treatment mean differences and for situations in which the researcher expects the null hypothesis of no treatment differences to represent the true state of nature. Biological researchers should be able to define the experimental unit in every biological research scenario and should be able to replicate treatments at the level of the experimental unit. Power analyses can be extremely effective to provide researchers with an objective mechanism to choose the number of replicates to balance statistical and logistical concerns. Power analyses can also be used to efficiently allocate resources among various types or forms of replication, including locations, years, and sampling units, among others.

“Everything is different from everything else,” so it was always said by Prof. Frank N. Martin of the University of Minnesota to the students in his introductory statistics courses. By far, most of the comparative experiments conducted in biological research are designed to detect differences between treatments or systems. As such, researchers create treatment designs in which the individual treatments are viewed as “different” from each other and likely to result in rejection of the null hypothesis for measurement variables of interest to the researcher. Power is simply the probability of correctly rejecting a null hypothesis that two or more treatment means are equal to each other when in fact they are different—in nontechnical terms, the likelihood of “getting it right” with a high degree of confidence, for example, $\alpha = 0.05$ or 95% confidence. The trick, or secret, according to Prof. Martin, is to design experiments that are unlikely to fail in this regard.

Conversely, we occasionally find ourselves in the situation in which our null hypothesis that there are no treatment differences is exactly what we expect to happen. For example, in a breeding and selection program, we often test whether promising new candidate cultivars have disease resistance similar to that of a commercial reference cultivar. In these situations it is imperative that powerful

M.D. Casler, USDA-ARS, US Dairy Forage Research Center, 1925 Linden Dr., Madison, WI 53706-1108 (mdcasler@wisc.edu, michael.casler@ars.usda.gov).

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5585 Guilford Road, Madison, WI 53711-5801, USA.

experiments be designed so that researchers can conclude with reasonable certainty that the true treatment means actually do not differ from each other. Consider a simple experimental situation in which the experiment is completed, the analysis is finished, and the final result is a failure to reject the null hypothesis of no differences among treatment means. What can you conclude? At this point, there are several possible explanations:

1. The treatments are truly not different from each other for the variable of interest.
2. The experimental design failed to account for spatial or temporal variation among experimental units.
3. There were inconsistencies, incongruities, or disturbances introduced during the course of the experiment that compromised the results.
4. There was insufficient replication or the type of replication was poorly chosen.

Explanation 1 relates to the choice of treatments, while the latter three relate to the design and conduct of the experiment, that is, its power. If there is any doubt about the quality, integrity, or power of the design, then it is virtually impossible to develop a definitive conclusion regarding the true differences among treatment means, so one cannot choose Explanation 1. The goal in this situation is to separate Explanation 1 from Explanations 2 through 4. Power, which is $1 - \text{the probability of a Type 2 error } (\beta)$ (see Chapter 1, Garland–Campbell, 2018), is decreased when any or all of Explanations 2 through 4 occur. Thus, when an experiment fails to detect treatment differences at a desired Type 1 error rate *and* power is low, the reasons for lack of significant differences cannot be clarified. Was the lack of significant differences due to low power, resulting from a design deficiency, a disruption or disturbance, or a lack of differences being the true state of nature?

One goal in designing biological experiments should be to reduce this form of doubt in a reasonable and economical manner, in other words, to create powerful experiments with low probability of a Type 2 error (see Chapter 1, Garland–Campbell, 2018). Blocking, randomization, and spatial analysis are important mechanisms for designing powerful experiments (see Chapters 3, Casler, 2018, and Chapter 12, Burgueño, 2018). Adherence to proper experimental methods, maintaining careful control over experimental conditions and protocols, and taking good notes are all mechanisms to help avoid problems related to Explanation 4 above. Replication is one of the most important factors that can be manipulated by the researcher to enhance power of biological experiments. Most of this chapter will deal with number and type of replicates as they influence power.

Concepts of Replication

What is an Experimental Unit?

First and foremost, proper replication demands that treatments be replicated at the scale of the experimental unit (Casler, 2015). **The experimental unit is the smallest unit to which a treatment is applied. Experimental error can only be estimated when treatments are applied to multiple and independent experimental units and data are collected on each individual experimental unit.** Replicate observations must occur at a spatial and temporal scale to match the application of treatments (Quinn

and Keough, 2002). Replication at the appropriate scale is essential because of the inherent variability that exists within biological systems and to avoid confounding treatment differences with other factors that may vary among experimental units.

A common mistake in designing biological experiments occurs due to pseudo-replication, in which treatments are not truly replicated. Figure 1A represents this design, with each column representing a single treatment or, alternatively, each treatment represents a “block” of samples. In this case, each treatment is applied to only one experimental unit. Multiple samples or data points may be generated within each column, but no matter how many samples of each treatment are collected, columns are completely confounded with treatments. Common examples include: (i) large agricultural or biological engineering experiments designed to compare two or more harvesting systems; (ii) harvesting, storage, or logistical experiments in which each system is assigned one physical space or site; (iii) ecological experiments designed to compare different in situ ecosystems or habitats; (iv) glasshouse or growth chamber experiments designed to compare different environmental conditions; or (v) laboratory experiments in which duplicates or triplicates of each treatment are submitted for analysis as a group. In the latter case, any temporal variation that exists in the laboratory analysis will be confounded with treatments. The lack of randomization and replication are the missing key elements that could be used to avoid this problem. Many laboratory experiments, for example, those dealing with conversion of biomass to bioenergy, simply state that samples were run in duplicate or triplicate, providing no additional information on the randomization or arrangement of samples in time (Casler et al., 2015).

In each of the five examples above (Fig. 1A), the only form of replication is to collect multiple samples within each experimental unit (or column in Fig. 1A), for

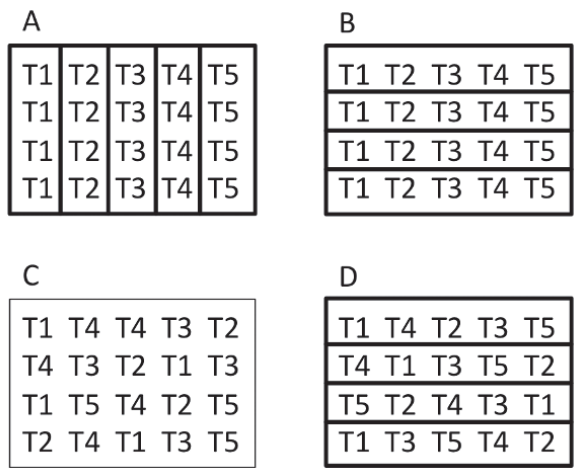


FIG. 1. Four possible designs of an experiment with five treatments (T1-T5) and four observations per treatment. (A) The experimental unit is a column with one experimental unit per treatment and four observations made on each experimental unit. (B) There are four experimental units per treatment, but they are not randomized. This is a common design in benchtop experiments. (C) There are four independent experimental units per treatment, all randomized together in a completely randomized design. (D) There are four independent experimental units per treatment, grouped into four blocks, randomized within each block in a randomized complete block design.

example, multiple hay bales within a field, multiple pots within a room or chamber, or multiple flasks in the laboratory. For many of these experiments, researchers have made the a priori decision that replication at the experimental unit scale is too expensive or time consuming. The consequences of this decision are twofold: (i) **treatments are confounded with experimental units**, such that all physical and biological effects that vary among experimental units are intermixed with treatment effects, and (ii) **error variances or residual terms are underestimated** (Casler, 2013; Casler et al., 2015). The direct consequence of confounding is to mix treatment effects with environmental effects, such as field plots, glasshouse benches, animals, etc. The direct consequence of underestimating the residual variance is that the true Type 1 error rate will be $> \alpha$ by an unknown amount. This is a very undesirable result that is quite common in some research areas (Casler et al., 2015).

Consider hay storage experiments as an example. A common treatment effect is the storage system, which typically includes variations on the following themes: indoor vs. outdoor; grass, gravel, or concrete pad base; or different types of coverings. Numerous hay bales are placed within each of the storage conditions for a certain period of time, without replication of the storage conditions. Each storage “site” is an experimental unit, and each hay bale is a sampling unit as shown in Fig. 1A, where columns correspond to storage sites and treatment and each alphanumeric character corresponds to one hay bale. All differences between storage sites are confounded with treatments. Researchers implicitly assume that the treatment effect itself accounts for all variation among experimental units. Proper replication at the experimental unit scale would involve: (i) two or more independent storage sites for each defined treatment (i.e., two concrete pads, two gravel pads, two indoor sheds, etc.) or (ii) complete repetition of the experiment at another location, during another harvest period, or in another year. A close colleague of mine, a chemist, often jokes, “Why ruin a good experiment by trying to replicate it?” This philosophy seems to be prevalent in quite a few research areas that deal with biological substrates, which are subject to sources of variation that are frequently ignored, misunderstood, or underestimated by non-biologists. As an example, this is a common problem in benchtop digestion experiments of biomass conversion to bioenergy (Casler et al., 2015).

Figure 1B represents one step in the right direction: avoiding the confounding associated with grouping together all units of one treatment. Treatments are “blocked” in either space or time so that the variation associated with the four row blocks can be removed from the residual variance. However, the lack of randomization confounds treatment variation with either spatial or temporal variation within each row block, leading to improper estimates of treatment means and probable underestimation of the residual variance. The design in Fig. 1B is frequently employed for benchtop or laboratory experiments by researchers who are not well versed in the importance and nuances of randomization and replication. For example, in forage, grain, or biomass quality analyses, all tissue samples must be processed through a grinder or mill with blades that become progressively dull over time. As blades lose their sharpness, particle-size distribution gradually changes, resulting in predictable changes in the ability of laboratory procedures to extract or accurately assay chemical compounds. The systematic nature in which treatments are

processed through the grinding mill results in confounding of treatments with blade sharpness and particle size, leading to inaccurate estimates of treatment differences.

Proper replication and randomization, as shown in Fig. 1C and 1D, are critical steps toward eliminating this problem for laboratory assays (Casler, 1992). These two figures represent two different and relatively simple designs, the completely randomized design (1C) and the randomized complete block design (1D), which are described in more detail in Chapter 3 (Casler, 2018). The design in 1C should be employed in situations for which the researcher expects limited or no potential for spatial or temporal variation. This has usually been established by the researcher from previous experiments. Conversely, design 1D should be employed if previous experiments have demonstrated any potential for spatial or temporal variation, allowing the row blocks to remove these sources of variation.

Replication on Multiple Scales

Experimental replication can occur in many forms, which can be classified into four basic levels within the experiment: (i) the experimental unit, as discussed above; (ii) replication of the entire experiment, as with multiple locations and/or years; (iii) sampling at one or more levels within experimental units; or (iv) repeated measures. The experimental unit is the most critical and should be given first and primary attention. Once the basic experiment has been designed with multiple and independent experimental units per treatment, then consideration can be given to replication at additional scales or levels. Many experiments are repeated at multiple locations or years, largely to broaden inferences by evaluating treatment differences across a range of environmental conditions, but also allowing estimation of treatment $\times$ environment interactions. In some cases, data cannot be collected on the entire experimental unit, requiring multiple samples to be collected within each experimental unit, each sampling unit representing a subset of the experimental unit. Lastly, some variables are measured repeatedly over time during the course of the experiment. If analyzed together in a single ANOVA, these factors are treated as repeated measures (see Chapter 10, Gezan and Carvalho, 2018). Repeated measures can be construed to represent true replication of treatments only when the autocorrelation between repeated measures is zero, which is probably quite rare.

Mixed models analysis provides estimates of random effects that impact a variable of interest. For most mixed model experiments, those in which fixed treatment effects are of primary interest, estimates of random effects are not of specific interest. However, estimates of random effects from previous experiments provide a mechanism for researchers to conduct retrospective exercises in resource allocation. For example, consider an experimental situation in which field studies repeated at multiple locations are an annual activity, such as uniform cultivar evaluations (e.g., Conaghan et al., 2008; Yaklich et al., 2002). Decisions regarding the relative numbers of locations, years, and replicates are critical for designing efficient future experiments, especially if there is a perceived deficiency or inefficiency in the existing design, or some desire arises to improve on the existing design. An ANOVA or mixed models analysis provides information that can be used to develop these inferences. Estimates of random effects from previous experiments are a frequently untapped resource, especially if they have sufficient degrees of freedom to be reliable estimates of true values. Methodology described in the next section can be adapted to

predict the impact of changes in resource allocation and design factors on the power of hypothesis tests for future biological experiments.

Replication and Power

The number of replicates required for any given experiment is a function of four variables: (i) the desired Type 1 error rate, (ii) the desired Type 2 error rate, (iii) the residual variance, and (iv) the desired detection level or the difference between treatment means to be detected. The relationships among these four factors can be evaluated in a relatively simple formula that applies to a two-sample t test. The minimum number of replicates required to detect a specified treatment mean difference (n) is

$$n = (t_{\alpha/2} + t_{\beta})^2 \sigma_D^2 / \delta^2$$

where the two t values are Student's t for the desired Type 1 and Type 2 error rates (α and β , respectively), σ_D^2 = the variance of differences or $2\sigma^2$, and δ = the desired detection level between treatment means (Steel et al., 1996, p. 123). Accurate estimates of n require a small number of iterations, because t is dependent on degrees of freedom, which are, in turn, dependent on n . Alternatively, if the number of treatments is sufficiently large and error degrees of freedom can be assumed to be > 30 , $Z_{\alpha/2}$ and Z_{β} can be used in place of the two t values.

The formula above illustrates the impact of four factors on the number of replicates for a given experiment. Reductions in either or both Type 1 and Type 2 error rates require an increase in the number of replicates. Thus, increasing the number of replicates results in increased power of hypothesis tests. Larger experimental error or residual variance leads to greater requirements for replication, all other things being equal. Finally, the smaller the desired detection level (δ), the more replicates are required, all other things being equal.

Gbur et al. (2012, Chapter 7) developed a more efficient and generalized manner of solving this problem, illustrating this approach for numerous scenarios. Their approach takes into account all four variables in the above formula, but it can be generalized to include multiple sources of variation and different designs that may optimize allocation of resources across different forms of replication. Five steps are required to implement the probability distribution method to estimate power of a future hypothetical experiment:

1. Obtain an estimate of the experimental error variance and variances of any other pertinent random factors from previous experiments conducted in the appropriate field or laboratory setting or from the literature.
2. Identify the appropriate distribution of the variable of interest, for example, normal, binomial, binary, etc.
3. Determine the p value that will be used to set detection limits [α = probability of a Type 1 error under the null hypothesis] and the minimum difference between treatments to be detected (δ).
4. Choose an experimental design structure that includes all desired blocking arrangements and one or more levels of replication [see Chapter 3 (Casler, 2018) and the previous section of this chapter for more details].

5. Create a representative data set that matches the desired experimental design structure and contains two “dummy” treatments with constant values across all experimental and observational units of the data set. To create such a representative data set, one must choose a single value for all the replication factors in the hypothetical experiment.

Exercise 1 shows SAS code to accomplish a single power analysis in three coding steps: (i) create a representative data set for a completely randomized design with two treatments, four experimental units per treatment (replicates), and two observational units per experimental unit; (ii) use generalized linear mixed models (Proc GLIMMIX) to compute the noncentrality parameter of the F distribution with the appropriate degrees of freedom under the alternative hypothesis that the two treatments have difference δ ($100 - 95 = 5$ in this example); and (iii) compute power from the noncentrality parameter. The SAS code in Exercise 1 computes power for a test with the following parameters and assumptions: treatment means = 95 and 100, variance components = 5 and 10 (experimental error and sampling error, respectively), $r = 4$ replicates, $s = 2$ observational units per experimental unit, and the assumption of normally distributed errors. All of these values can be changed within the SAS code, including the Type 1 error rate ($\alpha = 0.05$ in this example). In so doing, it is relatively easy to study the direct impact of design decisions on the predicted power of a future experiment. The predicted power for a future experiment with this design is 0.47.

Exercise 2 shows a modified version of this SAS code that contains a macro, automating the code to allow estimation of power for numerous experimental design parameterizations. This code estimates power for this design with number of replicates ranging from 4 to 6 and number of observations per experimental unit ranging from 2 to 3, with a wider range of values displayed in Fig. 2. Such a graphical display easily allows any researcher to choose one of multiple design arrangements that meet the expected target for power, which is often set at 0.8 (80%). For example, the results of this exercise easily illustrate a general principle of resource allocation, that the largest sized replication factors (experimental units in this case) have the greatest direct impact on improving power of future experiments. In the case of Fig. 2, using five different values for number of observational units per experimental unit, any one of five different scenarios can be chosen for use in future experiments, each with an expected power ≈ 0.8 .

Using this example and those provided by Gbur et al. (2012), power can be estimated for any imaginable design and sampling scenario, as illustrated in Fig. 2. Furthermore, once the decision is made and the experiment is completed, feedback can be obtained by a retrospective analysis of the experiment and its empirical detection limits. For example, Casler (1998) completed an extensive and expensive series of experiments that were designed to provide means and variance estimates to support a cultivar development program. The critical step involved direct comparison of numerous progeny populations created by various selection methods for which very small differences between means were expected. A power analysis was essential planning before designing the second experiment. Exercise 3 illustrates this power analysis and the process used to select the relatively unusual design employed by Casler (2008). While statistical theory, common sense, and the power

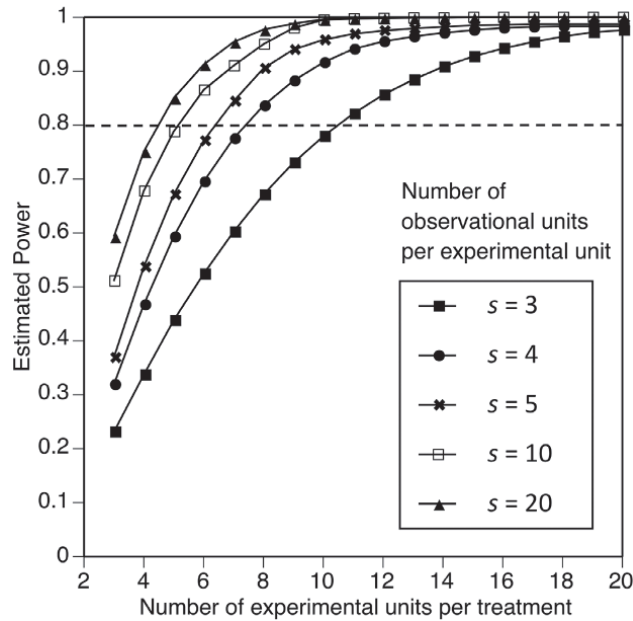


FIG. 2. Estimated power of a hypothesis test designed to detect a treatment difference of 5% of the mean with a Type 1 error rate of 0.05, variance component estimates of 5 and 10 (experimental and sampling errors, respectively), and varying numbers of experimental units and observational units ($s = 3$ to 20). The dashed line represents Power = 0.8 and illustrates that different replication and sampling scenarios can be created to provide similar expected results.

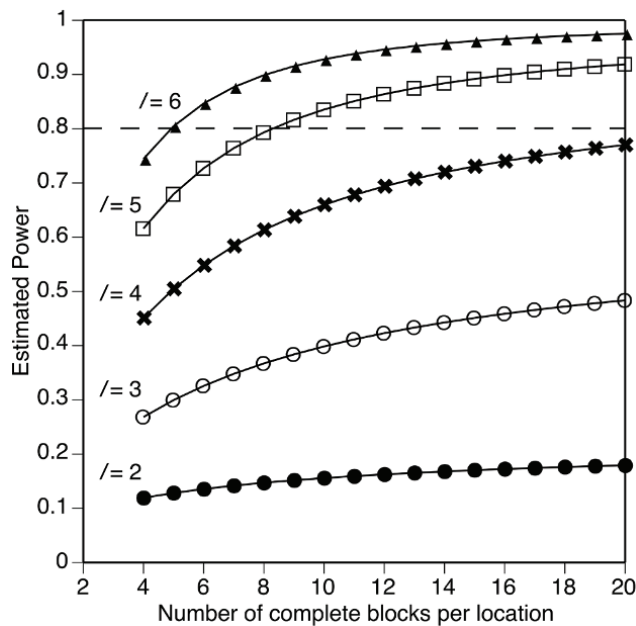


FIG. 3. Estimated power of a hypothesis test designed to detect a treatment difference of 5% of the mean with a Type 1 error rate of 0.05, variance component estimates of 0.02 and 0.2 (treatment $\times$ location interaction and experimental error, respectively) and varying numbers of locations ($l = 2$ to 6) and blocks. The dashed line represents Power = 0.8 and illustrates that different combinations of number of locations and replicates can be created to provide similar expected results.

analysis all suggested the need for four to six locations (Fig. 3), practical and logistical considerations led to the choice of three locations and 16 replicates per location (yes, 16 replicates per location—that's not a misprint), as a compromise, arranged in a randomized complete block design (Casler, 2008). The net result of this power exercise was a series of experiments with least significant differences (LSD) ranging from 2 to 3% of the mean and a high frequency of P values < 0.01 (Casler, 2008), that is, an extremely successful and satisfying result with LSD values about one-half of the target (5% of the mean; Exercise 3). This is an excellent practical example of using an a priori power analysis, combined with prior data and expectations, to alter normal replication strategies for a successful experimental outcome.

Conclusions

The potential economic, ecological, and sociological impacts of agricultural research are enormous compared with the low cost of research. However, public and private researchers who conduct experimental research are generally under severe budget constraints. Thus, the economic and emotional costs are high to researchers. As such, biological researchers should always follow the mantra, "failure is not an option." A failed experiment is one with generally high p values combined with low power, leaving the researcher with uncertain or equivocal conclusions: Are the treatments really not different from each other? Was my experimental design faulty due to poor planning and decision making? Was there some unknown and unseen disturbance that occurred to the experiment, causing errors to be inflated? Rarely can these questions be answered when p values are high and power of hypothesis tests is low, typically resulting in difficult-to-publish results and wasted time and money. To borrow an experimental design term, these causal explanations are confounded with each other when treatment effects are nonsignificant and power is low. It is generally impossible to assign cause to one or the other explanation. Designing experiments with predicted power levels as high as economically and logistically possible is critical for detecting expected differences among treatment means and for those experiments in which the null hypothesis of no treatment differences is the expected result.

Key Learning Points

Designing experiments with sufficient power to detect real treatment differences minimizes the chances that a researcher will be faced with inconclusive or equivocal results. Consider the following ideas and guidelines in planning future experiments.

- Be able to understand and define an experimental unit in any experimental situation in which you are involved.
- Strive to always replicate experiments at the level of the experimental unit and include additional levels or types of replication as necessary to improve precision or to broaden inferences.
- Always know the costs associated with various forms of replication so that you can easily conduct cost-benefit analyses of various replication strategies and make conscious decisions about which strategy gives the highest probability of success.

- Have a realistic idea of the treatment differences that you expect from any given experiment so that you can conduct a meaningful power analysis before making final decisions on the form and amount of replication.
- Do not become complacent with regard to replication and power. Conduct retrospective analyses of experiments, comparing your ability to detect differences among treatment means with your a priori power analyses, adjusting both experimental designs and replication strategies as needed to improve power of future experiments.

Exercises

1. Predict the power of a future hypothetical experiment using the probability distribution method.

PROBLEM. Guessing the appropriate number of blocks, replicates, or samples for any biological experiment is not a scientific method and is likely to result in experiments with low power of detecting treatment mean differences.

2. Simultaneously predict the power of several future hypothetical experiments using the probability distribution method and a SAS macro that allows for numerous power computations in a single run.

PROBLEM. There are potentially many satisfactory designs with sufficient predicted power to satisfy the goals of most researchers. A direct comparison of many possible designs is a desirable step toward choosing a statistically and logistically efficient design.

3. Design a new experiment to evaluate treatments that are expected to have very small mean differences, but for which it is critically important to be able to detect differences.

PROBLEM. The experiment to be designed is a follow-up experiment to that described by Casler (1998). Typical designs involve two or three locations and rarely more than four replicates per location. Least significant difference (LSD) values from these typical experiments were perceived to be too large, so a power analysis was conducted to identify the number of replicates and locations necessary to achieve a desired result.

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